

Soft-Sparse Scout TuRBO-TS

A trust-region Thompson sampler for expensive, high-dimensional optimization with heterogeneous coordinates and many far-apart good basins. It extends single-region trust-region Bayesian optimization along two independent axes (how a local step perturbs coordinates, and how the search reaches distant basins) and computes almost every internal constant from the problem itself, so the optimizer does not become a second tuning problem.

The exposition is layered so that the prose, the displayed equations, and the algorithm blocks are each self-supporting: the text motivates and explains, the equations define the objects precisely, and the algorithm blocks give the executable procedure. All three share the notation fixed in §0, so they can be read together or separately.

0. Notation

symbol	meaning
$f : [0, 1]^d \rightarrow \mathbb{R}$	objective; minimize; $f^* = \min f$
d, q, B	dimension, batch size, total evaluation budget
x_0	optional prior point (incumbent seed); may be absent
$\mathcal{D}_t = \{(u_i, y_i)\}_{i \leq n_t}$	observations after t asks; $y_i = f(u_i)$
$R = (c, \ell, v, \hat{y}, \hat{u}, \kappa)$	a region : center $c \in [0, 1]^d$, box side ℓ , visit count v , best value $\hat{y}$, best point $\hat{u}$, kind $\kappa \in \{\text{main}, \text{cand}\}$
$\mathcal{R}_t$	the region set; exactly one region has kind main
$\text{Box}(R) = \prod_j [c_j - \ell/2, c_j + \ell/2] \cap [0, 1]^d$	the trust region of R
$\alpha \in [0, 1]^d$	per-coordinate move weights (the local-move axis)
$M_m(\rho)$	mask law : a distribution on $[0, 1]^d$ for move m at density ρ
$\rho(v)$	target active-coordinate fraction at visit v
$g \sim \Pi(\cdot \mid S)$	a posterior function sample from a GP fit on data S
Σ	scout strategy (the escape axis): a policy on the state
S_y	robust scale of observed values (§5.3)
β, η	region-importance weights; δ novelty radius; c_0 Beta concentration

Prior information is (d, q, B, x_0) ; posterior information is $(\mathcal{D}_t, \mathcal{R}_t)$. A quantity is **derived** if it is a function of these, and a **choice** otherwise (§5.5).

1. The problem

Minimize an expensive black box in twenty to thirty dimensions. A decent prior configuration is usually known but is not the global optimum. The objective is locally smooth, so a local model is faithful. The coordinates are not interchangeable, and moving all of them at once at one scale tends to wreck a good configuration. Finally, the good configurations form many separated islands, and occasionally one distant island hides a much better narrow core that is only found by committing budget to it. Formally, solve

$$\min_{x \in [0, 1]^d} f(x), \quad d \in \{20, \dots, 30\},$$

from a batch-sequential oracle returning f at q points per round within a budget of B evaluations, under four assumptions:

- **(A1) local smoothness.** f is Lipschitz on each basin, so a GP posterior is faithful locally.
- **(A2) prior point.** an x_0 with $f(x_0) \ll \mathbb{E}_{x \sim \mathcal{U}} f(x)$ but $x_0 \notin \arg \min f$ (may be absent).
- **(A3) heterogeneity.** the active subspace of local improvement has effective dimension $d_{\text{eff}} \ll d$.
- **(A4) multi-basin.** $\{x : f(x) \leq f^* + \epsilon\} = \bigcup_k B_k$ with $\text{dist}(B_i, B_j) \gg r_{\text{local}}$, and some B_k contain a sub-basin of much lower value and much smaller radius.

These assumptions pull against each other. (A1)–(A2) demand hard local exploitation near x_0 ; (A4) demands a nonzero probability of leaving the current basin; (A3) demands that a local step move only a few coordinates. They are three separate requirements, met by three separate mechanisms: a local surrogate (§2), a scout (§3.3), and a coordinate mask (§3.1).

2. Base method and its structural limit

The base is trust-region Bayesian optimization, following TuRBO (Eriksson et al., 2019). Keep a box around the best point, fit a Gaussian-process surrogate inside it, propose the next points from that surrogate, and resize the box from feedback: grow it after a run of improvements, shrink it after a run of failures. The surrogate proposes points through a **Thompson-sampling acquisition**, which is TuRBO’s default choice for batch selection and the acquisition we use throughout.

It is worth being precise about how Thompson sampling is applied, because the version used here is the discretized, batch form rather than the textbook one-point rule. At each ask we do not optimize a posterior sample continuously. Instead we lay down a finite candidate pool inside the box (a quasirandom Sobol set), draw q independent function realizations from the GP posterior restricted to that pool, and take the pool-minimizer of each realization. The q minimizers are the batch. In symbols, fixing the single region R with center $c = \hat{u}$, the **candidate kernel** κ_R is the law of a uniform draw in the box and the **Thompson acquisition** TS_q is:

$$\kappa_R: X = c + r, \quad r \sim \mathcal{U}(\text{Box}(R) - c); \quad \text{TS}_q(R; \mathcal{D}) = \left\{ \arg \min_{x \in P} g^{(k)}(x) \right\}_{k=1}^q, \quad P \sim \kappa_R^{\otimes N}, \quad g^{(k)} \sim \Pi(\cdot \mid \mathcal{D}_R).$$

The box adapts through a success counter s and a failure counter ϕ :

$$\ell \leftarrow \min(\ell_{\max}, 2\ell) \text{ if } s \geq \text{succ_tol}; \quad \ell \leftarrow \max(\ell_{\min}, \ell/2) \text{ if } \phi \geq \text{fail_tol}.$$

Thompson sampling earns its place here for two reasons. First, it carries no acquisition trade-off constant. Expected improvement and upper-confidence-bound acquisitions both need a term or coefficient that trades exploration against exploitation, and in high dimension a global GP with such an acquisition tends to over-explore, which is the very failure that motivated TuRBO’s local trust region. A Thompson sample instead selects a point in proportion to its posterior probability of being the minimizer, so the exploration-exploitation balance is inherited from the surrogate’s own uncertainty rather than set by hand. Second, it gives batch diversity for free: because the q realizations are drawn independently, their minimizers naturally spread across the plausible optima without any constant-liar or penalization heuristic to keep a batch from collapsing onto one point. A stronger acquisition such as log-expected-improvement could be swapped in, but it would reintroduce a constant to tune and the per-batch diversity heuristic that Thompson sampling avoids, so we keep Thompson sampling as the base. Everything the method adds on top must preserve this knob-free property.

The limitation of the base is structural, not statistical. Once the box collapses onto one basin, a strictly better basin outside the box has probability of exactly zero of ever being sampled:

Proposition 1 (collapse). If $\text{Box}(R) \subseteq B_i$ and $\text{dist}(B_i, B_j) > \ell/2$ for a better basin B_j , then under the base method $\Pr[X \in B_j] = 0$ for the next proposal X .

Proof. $X \in \text{Box}(R)$ almost surely, so $\|X - c\|_\infty \leq \ell/2$. For any $z \in B_j$, $\|z - c\| \geq \text{dist}(B_i, B_j) > \ell/2 \geq \|X - c\|$, hence $X \neq z$ and $X \notin B_j$ a.s. $\square$

So far-basin discovery is impossible without an external channel, which is exactly what the scout axis supplies. A second, independent weakness is that the base kernel perturbs every coordinate at one shared width, which is the wrong step under heterogeneity (A3), and this is what the mask axis repairs.

Algorithm 1: base ask/tell.

```
ask(R, q):
    P ← N draws of c + U(Box(R) - c)           # candidate pool ~ κ_R
    G ← GP fit on region-local data D_R
    return [ argmin_{x∈P} g(x) for g in q posterior draws of G ] # TS_q

tell(R, y_batch):
    if best improved: s += 1; φ = 0 else: φ += 1; s = 0
    if s ≥ succ_tol: ℓ ← min(ℓ_max, 2ℓ); s = 0
    if φ ≥ fail_tol: ℓ ← max(ℓ_min, ℓ/2); φ = 0
```

3. Two orthogonal axes

The method is the base plus two independent choices: the **local move** (which coordinates a step perturbs) and the **scout** (how distant basins are reached). They are independent in the strict sense that any move composes with any scout.

3.1 The local move

A raw candidate r sits somewhere in the box. Rather than use it wholesale, each coordinate of r is mixed toward the region center c by a weight $\alpha_j \in [0, 1]$: weight one takes the box value, weight zero keeps the center, in between takes a partial step. A move is thus a law over the weight vector α , replacing the base kernel κ_R by the masked kernel

$$\kappa_R^m: \quad X = c + \text{diag}(\alpha) (r - c), \quad r \sim \mathcal{U}(\text{Box}(R)), \quad \alpha \sim M_m(\rho(v_R)).$$

Three laws $M_m(\rho)$, each with expected active mass $\mathbb{E} \sum_j \alpha_j = \rho d$:

$$\text{dense: } \alpha_j = 1; \quad \text{hard: } \alpha_j \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\rho); \quad \text{soft: } \alpha = \frac{\rho d}{\sum_j m_j} m, \quad m_j \stackrel{\text{iid}}{\sim} \text{Beta}(\rho c_0, (1 - \rho) c_0), \quad c_0 < 1.$$

Dense moves every coordinate (the base move). Hard moves a random subset fully and freezes the rest. Soft draws each weight from a *polarized* law: with $c_0 < 1$ most mass sits near 0 or 1, so it keeps the "mostly frozen" character of sparsity, yet it stays continuous, so it can also take a small coordinated step and its effective dimension varies smoothly with ρ . Because the three laws are one family, dense and hard are limiting cases of soft:

Proposition 2 (soft contains dense and hard). As $c_0 \rightarrow 0^+$ with mean ρ fixed, $M_{\text{soft}}(\rho) \Rightarrow M_{\text{hard}}(\rho)$; and $M_{\text{soft}}(1) = M_{\text{dense}}$ a.s.

Proof. With mean fixed at ρ and $c_0 \rightarrow 0^+$, the density $\propto x^{\rho c_0 - 1} (1 - x)^{(1 - \rho) c_0 - 1}$ diverges at both ends, concentrating mass on $\{0, 1\}$ with $\Pr[m_j = 1] \rightarrow \rho$, i.e. $\text{Bernoulli}(\rho)$; the renormalizer $\rightarrow \rho d / (\rho d) = 1$. At $\rho = 1$ both Beta parameters force $m_j = 1$. $\square$

Whether sparsity helps depends on the space, which is why the move is an explicit choice rather than a default:

Proposition 3 (sparsity is conditional). Let f have active set A of size d_{eff} (coordinates outside A do not change f). A ρ -sparse step leaves every irrelevant coordinate unmoved with probability $(1 - \rho)^{d - d_{\text{eff}}}$, versus 0 for a dense step. Hence when $d_{\text{eff}} \ll d$ (A3) a sparse step is far more likely to be supported on A , while at $d_{\text{eff}} = d$ freezing any coordinate only removes a usable direction.

Proof sketch. Improvement requires a favorable move within A ; perturbing coordinates outside A adds variance to the surrogate's credit assignment without changing f . Sparse masking raises the probability of a clean step supported on A precisely when A is small, and is pure loss of freedom when A is everything. $\square$

The move is a pure function of ρ and d , so new laws (top- k , learned masks) drop in without touching the rest of the method.

Algorithm 2: masked local candidate.

```
mask(m, ρ, d) → α ∈ [0, 1]^d:
  dense: α = 1
  hard : α = Bernoulli(ρ) per coord (force ≥1 active)
  soft : m = Beta(ρc0, (1-ρ)c0) per coord; α = clip(m · ρd/Σm, 0, 1)

local_candidate(R, m):
  r = U(Box(R)); α = mask(m, ρ(v_R), d); return c_R + α ⊙ (r - c_R)
```

3.2 Sparsity schedule

A fresh region does not yet know which coordinates matter, so it moves more of them; a mature region moves few, to polish without disturbing settled knobs. The active fraction decays with the region's visit count from a high start to a low floor,

$$\rho(v) = \rho_{\min} + (\rho_{\text{init}} - \rho_{\min}) e^{-v/\tau},$$

with endpoints and rate derived in §5.1.

3.3 The scout

The scout is the escape channel Proposition 1 makes mandatory. In practice a scout proposal rarely produces the incumbent improvement itself, since local steps do that, so its role is coverage insurance: keep the probability of reaching a distant useful basin above zero at low cost to local exploitation. A scout strategy is a triple $\Sigma = (\pi, \text{prom}, \text{alloc})$ of a **tick** (fire a scout this ask?), a **promotion** rule (turn an observation into a candidate region?), and an **allocation** (split the q slots across regions and moves):

$$\pi(\mathcal{R}_t, t) = [t \equiv 0 \pmod{p}] \vee [\text{no improvement} \geq \text{stag}],$$

$$\text{prom}(u, y, \mathcal{R}) = \left[\min_R \frac{\|u - c_R\|}{\sqrt{d}} \geq \delta \right] \wedge \left[y \leq Q_a(y) \vee y \leq \min y + c S_y \right].$$

The four strategies differ *only* in these three functions: whether they mine the log for candidates, how they promote, and how they split the batch. Those decisions are the definition of the strategy, not switches bolted onto a shared core:

Σ	mines log	allocation when $\pi = 1$	focus burst
none	–	q local on the single region	–
random	yes	$q-1$ local (round-robin over ranked $\mathcal{R}$) +1 far probe	–
sidecar	no	$q-s$ local on the protected main + s sidecar (candidate or probe)	–
switch	no	as sidecar, unless a focus episode is active	$q-1$ dense-in-candidate +1 main, for a bounded window

A far probe is the pool point farthest from all data, $u^* = \arg \max_{u \in P_{\text{far}}} \min_i \|u - u_i\|$. The escape channel is what makes distant basins reachable at all:

Proposition 4 (escape floor). Let μ_t be the probability a scout slot at ask t lands in $\bigcup_k B_k$, and ρ_t the fraction of ask- t slots that are scout slots. Then

$$\Pr[\text{some scout reaches a useful basin by ask } T] \geq 1 - \prod_{t=1}^T (1 - \rho_t \mu_t).$$

For $\Sigma = \text{none}$, $\rho_t \equiv 0$ and the bound is 0, consistent with Proposition 1. For any strategy with a periodic tick, $\rho_t \geq s/(pq) > 0$, so the floor is strictly positive and tends to 1 as $T \rightarrow \infty$ whenever $\inf_t \mu_t > 0$.

Proof. The complementary event "no scout ever lands" factorizes across ticks with per-ask non-landing probability at most $1 - \rho_t \mu_t$; take the product and complement. $\square$

Algorithm 3: scout ask/tell.

```

scout_ask( $\Sigma$ , R_set, q):
    tick = periodic(p) or stagnation(stag)
    if  $\Sigma = \text{switch}$  and focus_active:                # asymmetric burst
        return (q-1)×[dense-local in focus_cand] + 1×[local in main]
    if  $\Sigma = \text{none}$  or not tick:
        return round_robin_local(ranked(R_set), q)    # none: one region only
    return (q-s)×[local in main] + s×[candidate-or-far-probe]

scout_tell( $\Sigma$ , batch, y):
    for scouted (u,y): if prom(u,y): add_candidate(u)    # novelty  $\wedge$  value gate
    if  $\Sigma = \text{random}$ : mine_log_for_candidates()        # random only
    if  $\Sigma = \text{switch}$  and a candidate passes the focus gate: open/renew focus episode
    while |R_set| > max_regions: drop worst non-main region by I(R)

```

3.4 Candidate regions and importance

The multi-region strategies keep one main region on the incumbent and a few candidates. A candidate is created only if it is genuinely far from existing regions and passes the value gate; it first takes a few cheap probes, then fits its GP on *only its own local data*, so it explores its own basin instead of inheriting the incumbent's. This region-local training is the single most important implementation detail. When there are too many regions, the least valuable is dropped, but youth and distance are rewarded so a promising far region is not starved:

$$\mathcal{I}(R) = \hat{y}_R - \beta S_y \sqrt{\frac{\log(n_r+2)}{v_R+1}} - \eta S_y \nu(R), \quad \nu(R) = \min_{R' \neq R} \frac{\|c_R - c_{R'}\|}{\sqrt{d}}.$$

Lower $\mathcal{I}$ is kept; the worst non-main, non-warming region is dropped. Multiplying the uncertainty and coverage terms by the robust value scale S_y (§5.3) makes β, η pure numbers rather than task-dependent thresholds.

3.5 Why the scout must be conditional and asymmetric

On a deliberately hard structure (many look-alike basins, a few hiding narrow cores, a start planted off-core), three exploration designs separate cleanly. Serving all basins equally achieves the best coverage but the worst final value and never reaches a core: it over-explores and under-polishes. The sidecar strategy protects the local path and ties the strong baselines, but it still misses cores, because a strictly bounded side

channel never concentrates enough budget on any one candidate. The switch strategy is the first to reach a core, because it *concentrates* $q - 1$ slots on one candidate for a bounded window while keeping one protected main slot. The quantitative reason is a volume argument:

Reaching a narrow core of radius $r_c \ll \ell$ from outside requires $\Omega((\ell/r_c)^{d_{\text{eff}}})$ concentrated local samples (the shrinking-box volume ratio). A per-batch budget of s slots supplies $O(sT)$ total but only $O(s)$ *consecutive* samples on any one candidate, which is insufficient unless s is temporarily raised. The switch burst raises it to $q - 1$ for $\Theta(f_{\text{focus}}B/q)$ consecutive asks, which is the minimal change that makes the required concentration attainable while one main slot is protected.

The principle: exploration cheap and conditional; intensification temporary, bounded, and evidence-gated. An always-on burst would be the "serve every basin" failure again, so the switch is a conditional choice, appropriate when a far basin plausibly hides a better core, and not otherwise.

4. The method as one operator

One ask/tell loop. Update the robust value scale and each region's box and sparsity; if too few data, warm up; otherwise the scout strategy allocates the batch, the batch is evaluated, and the tell step adapts boxes, promotes scouted points, mines (random only), opens or closes a focus episode (switch only), and prunes regions. Writing the state as $z_t = (\mathcal{D}_t, \mathcal{R}_t, \theta_t)$ with counters θ , one step is the map Φ_Σ^m :

$$A_t = \bigsqcup_{(R, m') \in \text{alloc}(\mathcal{R}_t, q, \pi)} \text{TS}_1(R; \mathcal{D}_{R,t}) \text{ under } \kappa_R^{m'}, \quad Y_t = f(A_t),$$

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup (A_t, Y_t), \quad \mathcal{R}_{t+1} = (\text{prune} \circ \text{focus} \circ \text{mine} \circ \text{promote} \circ \text{adapt})(\mathcal{R}_t, A_t, Y_t),$$

where **adapt** applies the box dynamics of §2 per region, **promote**/**mine**/**focus** are the strategy functions of §3.3, and **prune** drops by $\mathcal{I}$ (§3.4). The method iterates $z_{t+1} = \Phi_\Sigma^m(z_t)$ from $z_0 = (\{(x_0, f(x_0))\}, \{R_{\text{main}}\}, \theta_0)$ until $|\mathcal{D}| \geq B$; Propositions 1 and 4 bound the reachability of Φ .

Algorithm 4: the full method.

```

input: mask m ∈ {dense, soft, hard}, scout Σ ∈ {none, random, sidecar, switch},
      problem (d, q, B), optional x0, risk
derive: n_init, pool N, max_data, (ρ_init, ρ_min, τ), (ℓ_init, ℓ_min, ℓ_max),
      succ_tol, fail_tol, p, stag, max_regions, δ, focus window # §5
init:   R_main ← region(x0 or none, ℓ_init); if x0: observe(x0)

while |D| < B:
  S_y ← robust_scale(y) # §5.3
  for R in R_set: adapt_box(R); ρ_R ← p(v_R) # §2, §3.2
  if |D| < n_init: ask quasirandom warmup; tell; continue
  batch, roles ← scout_ask(Σ, R_set, q) # §3.3; each slot:
  local_candidate(R, m')
  y ← f(batch); D ← D ∪ (batch, y)
  scout_tell(Σ, batch, y, roles) # promote / mine / focus / prune
return argmin over D

```

The following quantities enter the operator; the last column says whether each is computed or chosen.

quantity	value / rule	origin
$\rho(v), \rho_{\text{init}}, \rho_{\text{min}}, \tau$	§5.1	derived (d, B, q)
$\ell_{\text{init}}, \ell_{\text{min}}, \ell_{\text{max}}$	§5.2	derived (x_0, m, Σ)
succ_tol, fail_tol	§5.1	derived (d_{eff}, q)
n_init, N, max_data, max_regions, p, stag	§5.2	derived (d, q, B)
δ	§5.4	derived (cube geometry)
S_y	§5.3	posterior (y)
β, η, c_0	fixed	algorithm constants
Q_a, c , scout slots, focus fractions	§5.5	choice (Σ , risk)
m, Σ	the two axes	choice (declarations about the space)

5. Where the constants come from

Each rule below is derived from prior or posterior information and comes with the reason it is the right rule. The governing principle: compute from (d, q, B, x_0) or from the data; expose as a knob only a genuine choice (§5.5).

5.1 Effective dimension → sparsity and box patience

$$\rho_{\text{init}} = \text{clamp}\left(\frac{3}{\sqrt{d}}, 0.45, 0.70\right), \quad \rho_{\text{min}} = \text{clamp}\left(\frac{1}{\sqrt{d}}, 0.10, 0.25\right), \quad \tau = 0.25 \frac{B}{q}, \quad \text{succ_tol} = 3, \quad \text{fail_tol} = \max(4, \lceil \rho_{\text{init}} d / q \rceil).$$

By Proposition 3, sparsity pays when a step lands cleanly on the small active set. Under the standard assumption that the active set of an HPO landscape grows sublinearly, of order $\sqrt{d}$, choosing $\rho d \approx \sqrt{d}$, that is $\rho \approx 1/\sqrt{d}$, makes the expected number of moved coordinates match the active-set size: enough freedom to improve, few enough to keep steps clean. The floor $\rho_{\text{min}} \approx 0.20$ at $d = 25$ reproduces the density found best in direct sweeps, an independent check. The high start hedges early ignorance of the active set (§3.2). For **fail_tol**, each batch tests q perturbations of an order- $\rho_{\text{init}} d$ region, so declaring the box too large should take about one failed batch per effective direction, or $\rho_{\text{init}} d / q$ batches, which is neither trigger-happy nor sluggish.

5.2 Budget and dimension → sizes, cadences, focus window

$$\mathbf{n_init} = \max(2q, 8), \quad N = \max(256, 64q), \quad \mathbf{max_data} = \text{clamp}(4d, 64, 128),$$

$$\ell_{\text{init}} = \begin{cases} 0.35, & x_0 \text{ or masked/scouted} \\ 0.40, & \text{else} \end{cases} \quad \ell_{\text{min}} = 0.02, \quad \ell_{\text{max}} = \begin{cases} 0.90, & \text{scout or mask active} \\ 1.60, & \text{dense, no scout,} \end{cases}$$

$$p = \lceil s / (q f_{\text{scout}}) \rceil, \quad \mathbf{stag} = \max(4, \lceil 1.5 \mathbf{fail_tol} \rceil), \quad \mathbf{max_regions} = 1 + \min(3, \lfloor B/80 \rfloor), \quad \mathbf{focus_batches} = \lceil f_{\text{focus}} B / q \rceil.$$

N and **max_data** trade surrogate fidelity against the GP's $O(\mathbf{max_data}^3)$ cost; the cap grows with d but is bounded so cost stays flat. ℓ_{max} is small when a scout exists because global reach then comes from the scout (Proposition 4), not the box; a large box would only dilute local exploitation. The period p is set to hit a target scout fraction f_{scout} , so the ρ_t of Proposition 4 is a chosen budget rather than an accident. And **focus_batches** = $\Theta(f_{\text{focus}} B / q)$ is exactly the consecutive-concentration length §3.5 shows is needed to reach a narrow core.

5.3 Observed values → scale-free gates

$$S_y = \max(\text{IQR}(\mathbf{y}), 1.4826 \text{ MAD}(\mathbf{y}), \sigma_{\text{noise}}, \epsilon).$$

An absolute margin is meaningless across tasks whose scales differ by orders of magnitude. The factor 1.4826 MAD is the consistent Gaussian σ -estimator; IQR guards multimodal spreads; the outer max is conservative on heavy tails. Dividing every gate and the importance weights β, η by S_y turns them into pure numbers that transfer across tasks.

5.4 Cube geometry → novelty radius

$$\delta \approx 0.6 \sqrt{1/6} \approx 0.245.$$

Two uniform points in $[0, 1]^d$ have per-coordinate RMS distance $\sqrt{1/6} \approx 0.408$, independent of d . Requiring a candidate to be at least 60% of that typical distance from every region makes "novel" mean *materially separated*, not merely numerically distinct, so promotion creates genuinely different basins and the coverage term ν (§3.4) is on the same scale.

5.5 What stays a choice

Only what no prior or posterior can decide. There are three such discrete choices, all regime declarations about the landscape. The first is the move m : Proposition 3 shows the right move genuinely depends on whether the space is isotropic or heterogeneous, which only the user knows. The second is the scout Σ , which sets how much escape is warranted and how far to trust distant basins, and Proposition 1 versus §3.5 shows this is regime-dependent. The third is the trust-region update granularity $\in \{\text{batch}, \text{point}\}$: per-batch success/failure counting (standard TuRBO) shrinks the box slowly, which wins multimodal and basin-structured landscapes, while per-point counting shrinks it up to q times faster, which wins smooth convex valleys where aggressive convergence down one funnel is the goal. Neither dominates (the ablation is regime-split, like the mask), so it is a knob, not a derived constant, and the default is batch (the standard TuRBO semantics, and the safe choice for the many-basin target regime). A strategy's own behavioral constants, namely its promotion quantile Q_a , focus quantile, candidate-versus-probe probability, and slot count s , belong to the strategy, with validated defaults, not to the optimizer. The remaining surface is **budget** (which sharpens §5.2) and **risk**, a single word setting the margin coefficient c (conservative 0.10, balanced 0.25, aggressive 0.40), plus an optional explicit value or noise scale. In practice these choices are packaged as five named **presets** (§8), each the sweep-selected best config for one regime.

6. Components introduced here

The method reuses validated mechanisms but adds four small pieces not present verbatim in the prototypes, flagged so the reader knows what is new. First, treating dense and hard as the $c_0 \rightarrow 0$ and $p = 1$ limits of one Beta law (Proposition 2) is a consolidation, not a new behavior, since it is numerically identical at the endpoints. Second, the scale-free gates via S_y (§5.3) are applied uniformly to promotion, focus, and region importance; the prototypes used per-strategy absolute margins, and unifying them under one robust scale is new and makes the constants task-transferable. Third, the prior/posterior-derived schedule of §5 is new: the prototypes fixed these as constructor constants, and deriving them from (d, q, B, x_0) is what removes them from the user surface. Fourth, the trust-region update granularity is made an explicit axis with per-batch as the default: the prototypes counted trust-region success/failure per point, which fires the failure counter up to q times per batch and collapses the box before it can drift, badly hurting smooth funnels; per-batch counting matches standard TuRBO and what the **fail_tol** derivation assumes, and the two are exposed as a knob because the ablation is regime-split. The §5 derivations are arguments, not theorems, and where an assumption is heuristic, such as the active-set size of order $\sqrt{d}$, it is stated as such. Everything else (the masked kernel, the four scout strategies, region-local candidate training, importance-based pruning, the focus burst) is ported unchanged from the validated prototypes.

7. What the benchmarks show

What is tested. The three discrete axes (mask, scout, trust-region update) span $3 \times 4 \times 2 = 24$ configurations. All 24 were swept from scratch across every function of both problem families, ranked per function, and the per-function winners were grouped into six named **presets**, one per landscape regime. All three masks earn a preset, and the soft mask earns two, because it wins the smooth-anisotropic and the heterogeneous-multimodal regimes (Powell, Griewank, Michalewicz), which is the method's core contribution:

preset	mask	scout	tr_update	wins these functions
balanced (default)	hard	random	batch	Rastrigin; best all-round rank, escape insurance
rugged	hard	sidecar	batch	Ackley, Levy
smooth	dense	none	point	Rosenbrock, Dixon-Price
soft-smooth	soft	none	point	Powell, Griewank
heterogeneous	soft	none	batch	Michalewicz
multibasin	hard	switch	batch	many-basin family (both starts)

The six presets are compared against **standard, external optimizers**: CMA-ES (via the **cmes** library), GP-BO (global exact-GP surrogate with expected improvement), HEBO (warped-GP surrogate), and Sobol (quasirandom). All runs use one evaluation protocol (dimension $d = 25$, budget $B = 300$, batch $q = 4$); the standard suite is 5 seeds, the many-basin family 6. Lower is better throughout.

How the results are read. Each figure plots **best-so-far versus evaluations** (mean over seeds); lower and earlier is better. The standard suite is a 3×3 grid, **one panel per function** at that function's own axis scale (log where the range spans decades, with labelled decade gridlines; linear for the small-range or negative-valued functions), because the nine functions differ in scale by orders of magnitude and must not be pooled into one axis. Each panel's legend ranks the methods **by that function's own final value**, so the rank always refers to the single visible task and matches the curve that ends lowest in that panel. **A method keeps one fixed color across all panels** (presets solid, baselines dashed), so any method can be followed from task to task. The many-basin family is a single problem at one scale, shown as two panels (planted start x_0 , cold start **none**), each ranked by final value. Per-function raw final values are also tabulated below.

7.1 Standard synthetic suite (cold start, mean of 5 seeds)

Raw final value per function; **bold** = best method in the row (Michalewicz is a maximization objective, so more negative is better). Global-GP baselines are abbreviated where their values run to 10^5 – 10^6 .

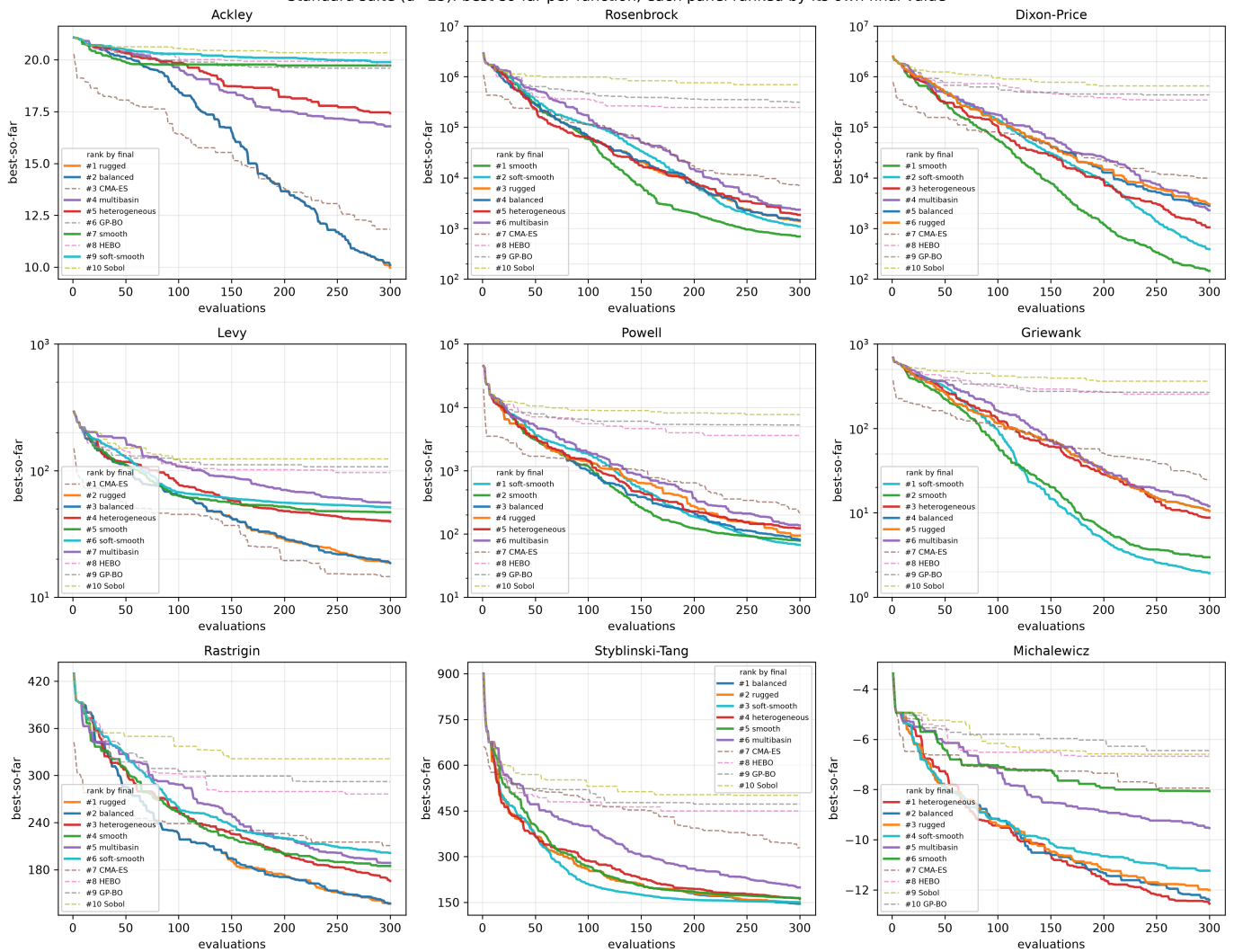
function	balanced	rugged	smooth	soft-smooth	heterog	multibas	CMA-ES	GP-BO	HEBO	Sobol
Ackley	10.1	10.0	19.7	19.9	17.4	16.8	11.8	19.6	19.7	20.3
Rosenbrock	1450	1390	695	1090	1850	2350	7080	3.1e5	2.5e5	6.9e5
Dixon-Price	2810	3020	145	390	1050	2280	9900	4.4e5	3.5e5	6.5e5
Powell	81	94	78	67	123	136	213	5260	3600	7690
Griewank	10.5	10.5	2.97	1.93	8.8	12.0	24.5	268	255	363
Levy	18.6	18.6	46.9	51.2	39.7	55.9	14.6	107	96.6	124
Rastrigin	137	137	185	201	166	189	211	292	276	321
Styblinski-Tang	146	150	164	151	161	199	329	472	450	501

function	balanced	rugged	smooth	soft-smooth	heterog	multibas	CMA-ES	GP-BO	HEBO	Sobol
Michalewicz	-12.4	-12.0	-8.1	-11.2	-12.5	-9.5	-7.9	-6.5	-6.7	-6.6
mean rank ↓	2.89	3.00	3.67	3.78	3.67	5.44	5.89	8.78	8.00	9.89

The mean-rank row (bottom of the table) is a single overall summary, though it is not the main point since each preset is built for a regime. All six presets rank ahead of every external optimizer (the best baseline, CMA-ES, is 5.89 versus 2.89 for balanced), and the presets take seven of the nine functions, with the two soft presets taking three of those seven and only Levy going to an external method.

The regime specialization is what actually matters, and it is visible per panel in the figure and per row in the table above. The soft mask, the method's core contribution, wins its regime cleanly: **soft-smooth** wins Powell and Griewank, the smooth-anisotropic functions where the graded partial step of the Beta mask pays off, and **heterogeneous** wins Michalewicz. **smooth** wins the two pure convex valleys (Rosenbrock, and Dixon-Price where it beats GP-BO by three orders of magnitude, 145 vs 4.4e5). On the rugged multimodal functions the hard mask leads: **rugged** wins Ackley, **balanced** wins Rastrigin and Styblinski-Tang, and only Levy goes to an external method (CMA-ES). This is exactly Proposition 3 in practice: soft for smooth-anisotropic, hard for rugged, dense for pure convex. The global-GP methods (GP-BO, HEBO) collapse in 25 dimensions, which is the failure mode the local trust-region approach exists to avoid.

Standard suite (d=25): best-so-far per function; each panel ranked by its own final value



7.2 Many-basin family (mean of 6 seeds)

gcore = fraction of runs reaching a narrow global core (regret ≤ 0.04); **ubasin** = distinct basins visited. Sorted by final within each start.

Planted start (x_0):

method	final	gcore
multibasin	0.133	0.17
smooth	0.146	0.00

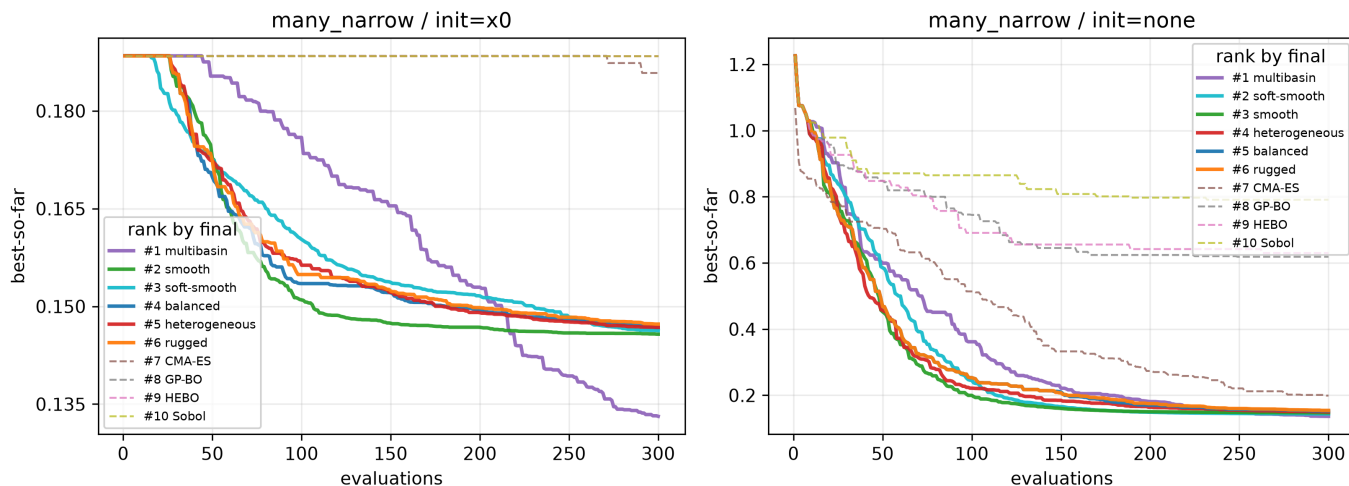
method	final	gcore
soft-smooth	0.146	0.00
balanced	0.147	0.00
heterogeneous	0.147	0.00
rugged	0.147	0.00
CMA-ES	0.186	0.00
GP-BO / HEBO / Sobol	0.188	0.00

Cold start (none):

method	final	gcore
multibasin	0.137	0.17
soft-smooth	0.144	0.00
smooth	0.147	0.00
heterogeneous	0.150	0.00
balanced	0.153	0.00
rugged	0.154	0.00
CMA-ES	0.199	0.00
GP-BO	0.619	0.00
Sobol	0.791	0.00

The **multibasin** preset is best on this family from both starts (0.133 / 0.137, reaching a narrow core in one run in six), and every preset dominates the global-GP baselines by a wide margin on the cold start (0.15 vs GP-BO's 0.62). The scouted presets (balanced, rugged) visit ~20 basins versus ~8 for the local ones, which is the coverage the scout axis is meant to buy.

Many-basin family (d=25): presets vs standard baselines



Preset selection methodology and the full 24-configuration sweep are in [preset-selection.md](#); the trust-region-update ablation behind the [tr_update](#) axis is in [tr_update-ablation.md](#). The many-basin family's global cores, though rare, are reachable by brute-force uniform sampling, so it is not used to claim superiority over random search. On landscapes where good regions are not stumble-into-able (the intended target and the standard suite), Sobol/quasirandom is last by a wide margin.

8. Choosing a configuration

The six benchmark-selected presets (§7) cover the distinct regimes, and each is the config ([mask](#), [scout](#), [tr_update](#)) that won its regime in the sweep:

preset	mask	scout	tr_update	use when
balanced (default)	hard	random	batch	the landscape is unknown; best all-round rank, escape insurance at low cost
rugged	hard	sidecar	batch	the objective is rugged / multimodal
smooth	dense	none	point	the objective is a smooth convex valley; fast box collapse converges fastest
soft-smooth	soft	none	point	smooth but the coordinates are anisotropic (a few sensitive, most inert)
heterogeneous	soft	none	batch	heterogeneous coordinates on a multimodal landscape (Proposition 3)
multibasin	hard	switch	batch	many far basins, one plausibly hiding a better narrow core (§3.5)

If nothing is known about the landscape, **balanced** is the safe default. The two soft presets carry the mask contribution: **soft-smooth** for a smooth but anisotropic objective, **heterogeneous** for an anisotropic multimodal one, the difference being only the trust-region update (per-point for the smooth case, per-batch for the multimodal one). The per-point presets (**smooth**, **soft-smooth**) collapse the box fast, which helps down a single funnel and hurts on rugged landscapes, so they are opt-in. Every result agrees on one point: a plain local trust-region search with no scout is the behavior to beat, not the method to ship, for the heterogeneous many-basin regime, but it is the right choice (the **smooth** or **soft-smooth** preset) when the landscape really is a single smooth funnel.

9. Limits

The switch advantage rests on a hard-case synthetic family and modest seed counts; it is a real effect on its target metric, narrow-core hits, but "wins one regime" is not "wins in general," and only the former is claimed. Selective moves are conditional by Proposition 3, which is why the move is an explicit axis. High observation noise degrades local and scout ranking, and broader-exploration methods can overtake; a noise-aware variant (duplicate-based noise detection gating repeated confirmation) is designed but not adopted, because always-on confirmation hurts the clean case. Vector-valued (multi-objective) scouting is a promising but unfinished extension. Adaptively switching the move or scout mid-run, and automatically selecting the initial configuration, are open directions pursued in a separate research note.